What Has TabPFN's Prior Learned?

Project update · DS-GA 3001: Visualization for Machine Learning · November 3, 2026

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Progress Since the Proposal

- **Foundation.** Built the shared repository: a data format for all runs, corrected metrics, nested synthetic datasets, a resumable sweep runner for local and Colab use, and a test suite.
- **Metric test.** On surfaces of known shape, the proposal's step score S gives a smooth ramp and a sharp edge the same score (1.00 at 0°, 0.71 at 45°). Our new jump share separates them (0.00 vs. 0.98 at 0°; 0.00 vs. 0.96 at 45°).
- **Classic rerun.** 84 runs on the new pipeline, at the pilot's size (4 datasets, n=60, 3 seeds), landed within 2.0 accuracy points and 0.02 axis alignment of the proposal's table. Mean jump share ranged from 0.00 (logistic regression) and 0.04 (GP) to 0.98 (gradient boosting) and 1.00 (k-NN).
- **Model access.** TabPFN v2 and 3.5 both ran on a Colab A100 without errors, taking about 3.2 s and 4.8 s per full 200×200 grid job. This single-problem test gives a rough estimate of 4 GPU hours for the planned runs.
- **Not yet complete.** Nhan's full classic sweep (8,960 runs) and real-dataset screen.

Response to Feedback

Feedback	Decision	What changed
1. Step score	Replaced	S is renamed axis alignment; jump share now measures sharpness. Jump share depends on its threshold (random forest: 0.94 at 0.01, 0.36 at 0.1), so we will fix the value before reading sweep results and report sensitivity.
2. Statistical strength	Accepted	10 seeds, with paired differences, 95% CIs, effect sizes and pre-set margins. Inconclusive results are reported as such.
3. Two datasets	Both options	About 6 datasets, assigned to tree- or smooth-favoring using classic models only, and framed as a case study. Fallback: the original two.
4. Classmate study	Removed	Replaced by 3–5 expert walkthroughs reported qualitatively, plus our worked examples.
5. Visualization claim	Accepted	D3 only; Streamlit removed. Claim: a slider shows one boundary evolving as context grows, and compare mode shows $\lvert pA - pB \rvert$, with a worked example of each. A static page cannot run TabPFN, so the slider uses precomputed nested contexts.
6. Model access	Checked	Go for H5, H6 and RQ3, on the result above. TabPFN runs remain on Kiet's Colab.

Scope and Plan

Completed: the foundation, metric test, classic rerun and model access check above. **Remaining:** all full sweeps, real-data tests, statistical analysis, the D3 interface, walkthroughs and the paper.

Dates	Kiet	Nhan
Nov 4–10	Freeze metrics, hypotheses and margins	Finish classic sweep and dataset screen; D3 canvas and controls
Nov 11–17	TabPFN synthetic sweep; RQ1 and RQ3 analysis; paper skeleton	Small multiples and metrics panel
Nov 18–24	TabPFN real-data runs; RQ2 analysis	Real-data pipeline and baselines; slider and compare mode
Nov 25–Dec 1	Full paper draft; walkthroughs	Story mode; walkthroughs
Dec 2–8	Freeze results Dec 5; revise paper	Clean-clone reproducibility test; polish interface; revise paper
Dec 9–12	Final fixes; submit by Dec 11	Final fixes; Dec 12 is buffer

If time runs short, we cut in order: ensemble-size ablation, noise-column test, H6, the additional datasets, story mode.

GitHub Repository

https://github.com/KNguyen37/tabpfn-inductive-bias (public; the README lists how to reproduce current results)